An explicit local zero-count bound for the Riemann heat deformation

Drafted for Edward Baker

17 October 2026
Revised working draft; round-two extension awaiting cross-model review

Abstract

Polymath established local zero counts of order $\log X$ for the Riemann heat deformation, uniformly for positive heat times at most $1/2$. We extract explicit constants in the physical coordinate $H_0(z) = \xi((1+iz)/2)/8$. The number of zeros with $X < \operatorname{Re} z \le X+1$, counted with multiplicity, is at most $(9/10)\log(X+2)+6$ for $0 < t \le 1/2$ and $X \ge 10^5$. For $X \ge 2\cdot 10^6$, the additive constant improves to $11/2$. The proof makes an off-axis comparison uniform in time, extends a Riemann–Siegel remainder estimate to both required real-part ranges, and combines Schwarz–Pick with a conjugate-pair Poisson-kernel bound. A separate height-selected remainder is retained in the truncation estimate. All numerical constants and cutoffs are evaluated. The lower-height range, cumulative error constants and an application gain are not supplied. The new extension has same-model development checks only; no specialist human review has occurred. No claim about RH or an improved de Bruijn–Newman bound is made.

Draft status. This revision integrates the requested round-two continuation of 17 October 2026. The earlier manuscript, at repository commit `146b3f9`, received one cross-model LLM review (Claude Review 02, recorded in commit `001aaeb`), accepting its results as proved from named imports. That review does not cover the changed cutoff, sharper extraction or this integrated revision. Those changes have same-model checks only. The derivations, arithmetic and manuscript preparation used `gpt-6.1-sol` (Codex), reasoning effort `ultra`, with substantial LLM assistance. Compilation and layout checks are not mathematical review. No literature-priority claim is made. The author field identifies the intended recipient rather than asserting endorsement.

1 Main result and relation to existing estimates

Let

$$\Phi(u) = \sum_{n\ge 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \tag{1.1}$$

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\,\mathrm{d}u. \tag{1.2}$$

The normalization is

$$H_0(z) = \tfrac{1}{8}\xi((1+iz)/2), \qquad \xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \partial_t H_t = -\partial_z^2 H_t. \tag{1.3}$$

Thus a zeta ordinate of magnitude T corresponds to physical coordinate $x = 2T$. Every logarithm without a capital Log is natural. For $X \ge 0$, define

$$N_t(X) = \sum_{0<\operatorname{Re}\rho\le X} \operatorname{mult}(\rho), \qquad K_t(X) = N_t(X+1) - N_t(X), \tag{1.4}$$

where the sum is over zeros of H_t. A zero on the right wall of a window is included, one on its left wall is excluded, and a real double zero contributes two.

Theorem 1.1 (Explicit unit-window bound)**.** *For $0 < t \le 1/2$ and $X \ge 10^5$,*

$$K_t(X) \le \frac{2003}{2240}\log(X+2) + \frac{125}{21} \le \frac{9}{10}\log(X+2) + 6. \tag{1.5}$$

The count includes complex and multiple zeros; no reality, simplicity or separation hypothesis is imposed.

Corollary 1.2 (The original high-height domain)**.** *For $0 < t \le 1/2$ and $X \ge 2 \cdot 10^6$,*

$$K_t(X) \le \frac{2003}{2240}\log(X+2) + \frac{75}{14} \le \frac{9}{10}\log(X+2) + \frac{11}{2}. \tag{1.6}$$

Theorem 1.5(iv) of Polymath [1] already establishes a time-uniform estimate $K_t(X) = O(\log(2+X))$, together with a cumulative estimate. The purpose here is to evaluate constants for one local version on a declared domain. The cutoff is conservative and is not claimed optimal. The range $200 \le X < 10^5$ remains outside the bound. No consumer has supplied a numerical tolerance against which to measure an application gain.

At $X = 2 \cdot 10^6$, the previous exact allowance $(82123/72000)\log(X+2) + 574/27$ was about 37.81 (its rounded version $(6/5)\log(X+2) + 22$ was 39.41). The new rounded bounds are 19.06 and 18.56, respectively. These are upper allowances, not observed counts. The leading mean density $\log(X/(4\pi))/(4\pi)$ is about 0.953 there, so the result is still loose. Pairing conjugate zeros reduces the Poisson multiplier from 41/9 to 25/7. Together with Schwarz–Pick, this replaces the earlier additive allowance 574/27 by 125/21. These changes do not provide exact counts or a lower population.

For a time-zero benchmark, Bellotti–Wong [2, Corollary 1.6] give

$$N_\zeta(T+1) - N_\zeta(T) \le \frac{\log T}{2\pi} + 4.8405 \qquad (3 \le T+1 \le 30\,610\,046\,000).$$

At $T = 10^6$ the right side is less than 7.04. A unit physical window corresponds to a half-unit T window and is contained in this unit T window. This is context at $t = 0$, not an input to the proof or an estimate transferred to positive heat time.

The analytic input is a two-regime comparison. The nonvanishing carrier M_t is defined in Section 2.

Theorem 1.3 (Uniform off-axis comparison)**.** *For $0 < t \le 1/2$, $x \ge 10^5$ and $3 \le y \le 4$,*

$$\left|\frac{H_t(x+iy)}{M_t((1+y-ix)/2)} - 1\right| < \frac{5094127}{7350000} < \frac{347}{500} < \frac{7}{10}. \tag{1.7}$$

If in addition $x \ge 2 \cdot 10^6$, then

$$\left|\frac{H_t(x+iy)}{M_t((1+y-ix)/2)} - 1\right| < q_0 := \frac{96624791}{147000000} < \frac{48}{73} < \frac{33}{50}. \tag{1.8}$$

Section 6 proves both comparisons. Section 7 then obtains the counts using Schwarz–Pick and a positive logarithmic-derivative sum with conjugate pairs. No compactness constant or finite zero survey enters the argument.

2 Carriers and imported analytic estimates

On $\mathbb{C} \setminus (-\infty, 1]$, with principal logarithms, set

$$M_0(s) = \frac{s(s-1)}{16}\pi^{-s/2}\sqrt{2\pi}\exp\left\{\left(\frac{s}{2}-\frac{1}{2}\right)\operatorname{Log}\frac{s}{2}-\frac{s}{2}\right\}, \tag{2.1}$$

$$\alpha(s) = \frac{M_0'(s)}{M_0(s)} = \frac{1}{2s}+\frac{1}{s-1}+\frac{1}{2}\operatorname{Log}\frac{s}{2\pi}, \tag{2.2}$$

$$M_t(s) = e^{t\alpha(s)^2/4}M_0(s), \qquad \frac{M_t'}{M_t}(s) = \alpha(s)+\frac{t}{2}\alpha(s)\alpha'(s). \tag{2.3}$$

The functions have the expected conjugation symmetry and are nonzero on the domains used below. We use the conventions

$$T = \frac{x}{2}, \quad T' = T+\frac{\pi t}{8}, \quad Q = \frac{x}{4\pi}, \quad L = \log Q, \quad a = \sqrt{\frac{T'}{2\pi}}, \quad N = \lfloor a \rfloor, \tag{2.4}$$

and

$$s_+ = \frac{1+y-ix}{2}, \qquad s_- = \frac{1-y+ix}{2}, \qquad b_n^t = e^{t\log^2 n/4}, \qquad B_t(z) = M_t((1-iz)/2). \tag{2.5}$$

For $3 \le y \le 4$, the real parts of $s_\pm$ belong to $[-3/2, 5/2]$. Note that $a^2 = Q + t/16$.

We state the imported ingredients precisely. All equation and theorem numbers attributed to Polymath refer to arXiv version 2 of [1]. First, H_t is even and real entire of order one, $H_t(iy) > 0$ for real y, and its zeros lie in $|\operatorname{Im} z| \le 1$ for $0 \le t \le 1/2$; see its Section 3 and Theorem 3.2.

To describe its expansion (39), write

$$c(s) = \frac{s(s-1)}{16}\pi^{-s/2}\Gamma(s/2), \qquad r_{0,n}(s) = c(s)n^{-s},$$

and

$$R_{0,N}(s) = c(s)\int_{N\swarrow N+1}\frac{w^{-s}e^{i\pi w^2}}{e^{\pi i w}-e^{-\pi i w}}\,\mathrm{d}w.$$

The contour has direction $e^{5\pi i/4}$ and crosses $(N, N+1)$. For nonreal s, define $r_{t,n}$ and $R_{t,N}$ by Gaussian convolution in s, for example

$$R_{t,N}(s) = \frac{1}{\sqrt{\pi}}\int_{\mathbb{R}} R_{0,N}(s+\sqrt{t}\,v)e^{-v^2}\,\mathrm{d}v. \tag{2.6}$$

For $F^*(s) = \overline{F(\overline{s})}$, the exact formula is

$$\begin{aligned} H_t(z) = {} & \sum_{n=1}^{N} r_{t,n}(s_-) + \sum_{n=1}^{N} r_{t,n}^*(s_+) \\ & + R_{t,N}(s_-) + R_{t,N}^*(s_+). \end{aligned} \tag{2.7}$$

This formula and its contour shifts are imported from [1, (39)–(41)]; sub-Gaussian growth on the shifted lines justifies the Gaussian integrals.

Proposition 6.1 of [1] applies for every real σ, $T > 10$, $n \ge 1$ and $0 < t \le 1/2$. It gives

$$r_{t,n}(s) = M_t(s)b_n^t n^{-s-(t/2)\alpha(s)}(1+\theta_{t,n}(s)), \quad s = \sigma+iT, \tag{2.8}$$

$$|\theta_{t,n}(s)| \le \exp\left(\frac{(t^2/8)|\alpha(s)-\log n|^2 + t/4 + 1/6}{T-3.33}\right) - 1. \tag{2.9}$$

The r functions have conjugation symmetry, so the same estimates hold at the reflected arguments. We also import the general-real-part coefficient and remainder bounds of Proposition 6.2, including $|C_0(p)| \le 1/2$, and the explicit Stirling estimate of Lemma 5.1(v). We do not extend the final constants of its Proposition 6.3 beyond their stated real-part range. Section 4 rebuilds the needed remainder estimate instead. The coefficient and remainder bounds, including the C_0 estimate, originate with Arias de Reyna, and the Stirling estimate with Boyd, as cited by Polymath. We import those bounds through Polymath; their original papers have not been checked here.

3 Bounds for the ordinary terms

Lemma 3.1 (Heat weights and termwise error)**.** *For $x \ge 10^5$, $0 < t \le 1/2$, $-3/2 \le \sigma \le 5/2$ and $s = \sigma \pm iT$, uniformly for $1 \le n \le N$,*

$$\operatorname{Re}\alpha(s) \ge L/2 - 13/x^2, \tag{3.1}$$

$$|\alpha(s) - L/2 \mp i\pi/4| \le \frac{11}{2x} + \frac{25}{4x^2} < \frac{6}{x}, \tag{3.2}$$

$$W_n(s) := b_n^t n^{-(t/2)\operatorname{Re}\alpha(s)} \le 1, \qquad |\theta_{t,n}(s)| < \frac{1}{6000}. \tag{3.3}$$

For $x \ge 2 \cdot 10^6$, the termwise error improves to $1/50000$.

Proof. In (2.2), the real rational terms are bounded below by $-13/x^2$, and the log-modulus correction to $L/2$ is nonnegative. The inequalities $|\arctan(\sigma/T)| \le |\sigma|/T$ and $\log(1+\sigma^2/T^2) \le \sigma^2/T^2$, together with the rational terms, prove (3.2). Since $a^2 = Q + t/16$,

$$\log N \le L/2 + \frac{\pi}{16x}, \qquad |\alpha(s) - \log n| \le L/2 + 1 \le \log x.$$

The exact identity

$$W_n(s) = \exp\left\{\frac{t}{4}\log n(\log n - 2\operatorname{Re}\alpha(s))\right\} \tag{3.4}$$

and the bound $2\operatorname{Re}\alpha(s) - \log N \ge L/2 - \pi/(16x) - 26/x^2 > 0$ prove $W_n \le 1$. The heat coefficient in (3.4) is $t/4$.

Because $T - 3.33 \ge x/3$, the exponent in (2.9) is at most

$$u(x) = \frac{3(\log x)^2/32 + 7/8}{x}.$$

Both $(\log x)^2/x$ and $1/x$ decrease here. Since $\log 10^5 < 12$,

$$u(x) \le \frac{23}{160000} < \frac{1}{8}, \qquad e^{u(x)} - 1 \le \frac{8}{7}u(x) \le \frac{23}{140000} < \frac{1}{6000}.$$

For $x \ge 2 \cdot 10^6$, use $\log(2 \cdot 10^6) < 15$ to obtain $u(x) \le 703/(32 \cdot 2 \cdot 10^6)$ and $e^{u(x)} - 1 < 1/50000$. This proves both assertions. □

The dominant branch has real part $(1+y)/2 \ge 2$. Its leading normalized approximant is exactly 1, and its other terms total at most

$$\sum_{n=2}^{N} n^{-(1+y)/2} W_n(s_+) \le \zeta(2) - 1 \le \frac{95}{147}. \tag{3.5}$$

Here and below $\zeta(2) \le (22/7)^2/6 = 242/147$ is used as a convenient rational upper bound.

4 A remainder estimate for both real-part ranges

Lemma 4.1 (A remainder bound on an enlarged real-part interval)**.** *Let*

$$T \geq 50000, \qquad -\frac{3}{2} \leq \sigma \leq \frac{5}{2}, \qquad 0 < t \leq \frac{1}{2},$$

and set

$$T' = T + \frac{\pi t}{8}, \qquad a = \sqrt{\frac{T'}{2\pi}}, \qquad N = \lfloor a \rfloor.$$

For the remainder $R_{t,N}$ defined in [1, equations (38)–(41)], one has

$$\left|R_{t,N}(\sigma + iT)\right| < 2\exp\left(\frac{t\pi^2}{64}\right)|M_0(iT')|. \tag{4.1}$$

For $T \geq 10^6$, the prefactor 2 can be replaced by 1:

$$|R_{t,N}(\sigma + iT)| \leq e^{t\pi^2/64}|M_0(iT')|. \tag{4.2}$$

Proof. We use the general-real-part estimates of [1, Proposition 6.2], rather than extending the stated domain of its Proposition 6.3. Write

$$D = T - 3, \qquad p = 1 - 2(a - N), \qquad u = \sigma + \sqrt{t}\, v.$$

Expectation below refers to the density

$$\frac{1}{\sqrt{\pi t}}\exp\left(-\frac{(u-\sigma)^2}{t}\right) du.$$

Throughout the proof, $D \geq 49997$ and $a^2 > 87500/11 > 89^2$. The larger endpoint $a > 300$ is used only for the final high-height subcase.

The Gaussian envelope. Use the shift $\beta = \pi i/4$ in [1, equation (41)]. Its imaginary displacement is $\pi t/8$, so both the original and shifted contours remain in the upper half-plane. Apply Proposition 6.2 to $u + iT'$. After extracting the common carrier and its unit-modulus phase, the integrand is

$$e^{\eta(u)}\left(C_0(p) + \sum_{k=1}^{K_u}\frac{C_k(p,u)}{a^k} + RS_{K_u}(u + iT')\right),$$

where K_u will be chosen below. The extracted carrier has modulus

$$\exp\left(\frac{t\pi^2}{64}\right)|M_0(iT')|.$$

The explicit Stirling estimate in [1, Lemma 5.1(v)], the formula for α in its equation (9), and the derivative bound in its equation (43) give

$$\begin{aligned}|\eta(u)| &\leq \frac{u^2}{4(T'-3)} + \frac{3|u|}{2T'} + \frac{1}{6(T'-0.66)} \\ &\leq \frac{u^2 + 6|u| + 2/3}{4D} \leq \frac{u^2+1}{D}.\end{aligned} \tag{4.3}$$

Indeed, the horizontal Taylor segment from iT' to $u+iT'$ has constant imaginary part T', and $6|u| \le 3u^2+3$. Thus this estimate is valid for every real u; no boundedness assumption on u enters the Taylor argument.

For a real number b, put

$$J_b = \mathbb{E}\exp\left(\frac{u^2+1}{D} + bu\right), \qquad h = 1 - \frac{t}{D}.$$

Completing the square gives the exact formula

$$J_b = h^{-1/2}\exp\left(\frac{\sigma^2+1}{D} + b\sigma + \frac{t(\sigma/D + b/2)^2}{h}\right). \tag{4.4}$$

The simpler inequality $u^2 \le 2\sigma^2 + 2tv^2$ also gives

$$J_0 \le \exp\left(\frac{2\sigma^2+t+1}{T-6}\right) \le \exp\left(\frac{14}{T-6}\right) < \frac{4}{3}. \tag{4.5}$$

For $b = \log 9$ or $b = -\log 2$, the factor in (4.4) remaining after removal of $e^{b\sigma}$ is less than 2. To see this, use

$$h \ge \frac{999}{1000}, \qquad \frac{\sigma^2+1}{D} < \frac{1}{1000}, \qquad \frac{|\sigma|}{D} < \frac{1}{1000}, \qquad -\frac{1}{2}\log h < \frac{1}{1000}.$$

Since $\log 3 < 11/10$ and $\log\sqrt{2} < \log 3$, its logarithm is less than

$$\frac{2}{1000} + \frac{1}{2}\left(\frac{1101}{1000}\right)^2 \frac{1000}{999} < \frac{2}{3} < \log 2.$$

Coefficient summation and the fixed-index remainder. Choose

$$m = \lfloor 2a^2 \rfloor, \qquad K_u = \begin{cases} 1, & u \ge 0, \\ \max(\lfloor -u \rfloor + 3, m), & u < 0. \end{cases}$$

When $u < 0$, this choice satisfies $K_u + u \ge 2$, as required by [1, equation (57)]. Define

$$S(u) = \sum_{k=1}^{K_u} \frac{|C_k(p,u)|}{a^k} + |RS_{K_u}(u+iT')|, \qquad E_a = a^{-1}e^{-a^2/2}.$$

The following pointwise bound includes every remainder term:

$$\begin{aligned} S(u) \le{}& \frac{0.2\,9^u}{a-0.865} + \frac{0.029\,2^{-u}}{a-0.353} \\ &+ \frac{2^{-u}}{2} \sum_{2a^2 \le k \le -u+4} \frac{1.1^k \Gamma(k/2)}{a^k} + E_a. \end{aligned} \tag{4.6}$$

All sums over k have integer indices; an empty sum is zero.

Here are the details of the summation. For $u \ge 0$, the bounds in [1, equations (54) and (56)] give

$$S(u) \le \frac{0.200\,9^u}{a} + \frac{0.173\,2^{3u/2}}{a^2} \le \frac{0.2\,9^u}{a-0.865}.$$

The value $u = 0$ can be assigned by continuity and has no effect on the integrals. For $u < 0$, the coefficients with $k \le m$ satisfy

$$\sum_{k=1}^{m} \frac{|C_k(p,u)|}{a^k} \le 0.036\, 2^{-u} \left(\frac{0.445\sqrt{\pi}}{a} + \frac{0.247}{a^2} + \frac{0.098}{a^3} \right) \le \frac{0.029\, 2^{-u}}{a - 0.353}.$$

This follows from [1, equation (55)] by separating even and odd indices and using

$$\frac{\Gamma((k+2)/2)}{a^{k+2}} = \frac{k}{2a^2} \frac{\Gamma(k/2)}{a^k} \le \frac{\Gamma(k/2)}{a^k} \qquad (k \le 2a^2).$$

The constants bound the corresponding geometric sums. The final compression remains valid at the smaller endpoint, since

$$.036 \left(.445 \frac{71}{40} + \frac{.247}{89} + \frac{.098}{89^2} \right) < .029, \qquad \sqrt{\pi} < 71/40.$$

If $K_u > m$, every coefficient beyond m has index greater than $2a^2$. Moreover the remainder index satisfies

$$K_u + 1 = \lfloor -u \rfloor + 4 > 2a^2, \qquad K_u + 1 \le -u + 4.$$

The bounds in equations (55) and (57) of [1] therefore place these additional coefficients and the remainder inside the tail sum in (4.6).

If $K_u = m$, the remainder need not belong to that tail. To bound it, put $\nu = (m+1)/2$, so that $a^2 < \nu \le a^2 + 1/2$. Equation (57) and explicit real Stirling give

$$\begin{aligned} |RS_m(u + iT')| &\le \frac{1}{2}\Gamma(\nu)(1.1/a)^{2\nu} \\ &\le \frac{3}{2a} \exp\left(\nu \log(\nu/a^2) - \frac{4}{5}\nu \right) \\ &< \frac{5}{a} e^{-4a^2/5} \le E_a. \end{aligned}$$

We used $\Gamma(\nu) \le 3\nu^{\nu - 1/2} e^{-\nu}$, $\log(1.21) < 1/5$, $\log(\nu/a^2) \le 1/(2a^2)$, $\nu \log(\nu/a^2) \le 1/2 + 1/(4a^2) < 1$, and $3a^2/10 > 2 > \log 5$. These bounds hold already for $a > 89$. Thus the added E_a accounts for the fixed-index remainder even when the displayed tail range is empty. This proves (4.6) in every case.

Integration of the four contributions. Let $\delta_1, \dots, \delta_4$ denote the expectations of the four terms in (4.6), each multiplied by $\exp((u^2+1)/D)$. The moment bounds above give

$$\begin{aligned} \delta_1 + \delta_2 &< \frac{0.4\, 9^\sigma + 0.058\, 2^{-\sigma}}{a - 0.865} \\ &\le \frac{97.374}{a - 0.865}. \end{aligned} \tag{4.7}$$

Here $9^\sigma \le 243$ and $2^{-\sigma} < 3$. Retain this denominator: the shortcut $97.374/(a - .865) \le 98/a$ requires $a \ge 42385/313$ and is unavailable at $a = 89$.

In the third contribution, $2a^2 > 15842$, so enlarge the integer summation to $k \ge 14$. Its integration region has $u \le 4 - k \le -10$. Since $|\sigma| \le 5/2$,

$$(u - \sigma)^2 \ge \frac{9}{16} u^2, \qquad \frac{u^2+1}{D} - \frac{(u-\sigma)^2}{t} \le \frac{1}{D} - \frac{u^2}{2t}.$$

Furthermore,

$$\int_{-\infty}^{4-k} e^{-u^2/(2t)-u\log 2}\,du \le \frac{2t}{k-6}e^{-(k-4)^2+(k-4)\log 2}.$$

For

$$f(k) = 1.1^k\Gamma(k/2)e^{-(k-4)^2+(k-4)\log 2},$$

one has $f(14), f(15) < 10^{-30}$ and

$$\frac{f(k+2)}{f(k)} = 2.42k\,e^{-4k+12} < 1 \qquad (k \ge 14).$$

The initial bounds follow, for example, from $\Gamma(7) = 720$, $\Gamma(15/2) = 135135\sqrt{\pi}/128$, $\sqrt{\pi} < 2$, and $e > 8/3$. Summing the resulting geometric series in $1/a$ gives

$$\delta_3 \le \frac{4\cdot 10^{-30}}{a^{14}} < \frac{1}{a}. \tag{4.8}$$

The fourth contribution satisfies

$$\delta_4 = E_a J_0 < \frac{4}{3}E_a < \frac{1}{a}. \tag{4.9}$$

Finally, [1, equation (58)] gives $|C_0(p)| \le 1/2$. Taking absolute values in the shifted integral and using (4.3) therefore yields

$$\frac{|R_{t,N}(\sigma+iT)|}{e^{t\pi^2/64}|M_0(iT')|} \le \frac{1}{2}J_0 + \sum_{j=1}^{4}\delta_j < \frac{2}{3} + \frac{97.374}{a-.865} + \frac{2}{a}.$$

The last expression decreases in a and is less than

$$\frac{2}{3} + \frac{97.374}{89-.865} + \frac{2}{89} = \frac{42215698}{23532045} < \frac{9}{5} < 2.$$

This proves (4.1). For $T \ge 10^6$, one has $a > 300$, and now $97.374/(a-.865) \le 98/a$ is permitted. The same strict chain yields

$$\frac{|R_{t,N}(\sigma+iT)|}{e^{t\pi^2/64}|M_0(iT')|} < \frac{1}{2}\cdot\frac{4}{3} + \frac{100}{300} = 1,$$

proving (4.2). The endpoint on the right is an equality of rational allowances; strictness comes from the preceding analytic inequalities. □

5 Ratios of the carriers

Lemma 5.1 (Carrier comparisons)**.** *On the domain of Theorem 1.3,*

$$\left|\frac{M_t(s_-)}{M_t(s_+)}\right| < 2Q^{-y/2}, \tag{5.1}$$

$$e^{t\pi^2/64}\frac{|M_0(iT')|}{|M_t(s_+)|} \le Q^{-(1+y)/4}. \tag{5.2}$$

Proof. Put $A = (1+y)/2 \in [2, 5/2]$ and $C = 1 - A \in [-3/2, -1]$. For $-3/2 \le \sigma \le 5/2$, direct use of (2.2) gives

$$|\operatorname{Re}\alpha(\sigma+iT) - L/2| \le 85/(16T^2), \qquad |\operatorname{Im}\alpha(\sigma+iT) - \pi/4| \le 11/(4T),$$
$$|\alpha(\sigma+iT)| \le L/2+1, \qquad |\alpha'(\sigma+iT)| \le 1/(2T) + 3/(2T^2).$$

Integrating the real part of the logarithmic derivative in (2.3) from $C+iT$ to $A+iT$ gives

$$\log\left|\frac{M_t(s_-)}{M_t(s_+)}\right| \le -\frac{y}{2}L + \frac{85}{x^2} + (L/2+1)(1/x + 6/x^2) \le -\frac{y}{2}L + \frac{\log x}{x}.$$

For example, the final inequality follows from $L \le \log x$, $85/x \le 1/100$, $6/x \le 1/100$ and $\log x \ge 3$. Conjugation equates the denominator's modulus to that at $A+iT$. Since $\log x/x < 1/2$ and $e^{\log x/x} < 2$, this proves (5.1).

For the second inequality, set $G(\sigma,\tau) = \operatorname{Re}\log M_0(\sigma+i\tau)$. An exact horizontal integration gives

$$G(A,T) - G(0,T) - AL/2 = \tfrac{1}{4}\log(1+A^2/T^2) + \tfrac{1}{2}\log\frac{T^2+(A-1)^2}{T^2+1} + \tfrac{1}{4}\int_0^A \log(1+\sigma^2/T^2)\,\mathrm{d}\sigma \ge 0.$$

Every term is nonnegative because $A \ge 2$. On the imaginary axis,

$$\partial_\tau G(0,\tau) = -\frac{\pi}{4} + \frac{1}{2\tau} + \frac{\tau}{1+\tau^2} \le -\frac{\pi}{4} + \frac{3}{2T} \qquad (\tau \ge T).$$

At $A+iT$, the exact formula for α gives $\operatorname{Re}\alpha \ge L/2$ and $0 \le \operatorname{Im}\alpha \le \pi/4$. Thus

$$\operatorname{Re}(\alpha(A+iT)^2) \ge L^2/4 - \pi^2/16.$$

Combining these estimates, including the displacement $T' - T = \pi t/8$, yields

$$e^{t\pi^2/64}\frac{|M_0(iT')|}{|M_t(s_+)|} \le Q^{-A/2}\exp\left(-\frac{tL^2}{16} + \frac{3\pi t}{8x}\right). \tag{5.3}$$

The exponent is nonpositive because $L > 1 > 6\pi/x$ on the stated domain. This proves (5.2). □

6 The complete off-axis budget

Proof of Theorem 1.3. Use $N = \lfloor\sqrt{Q + t/16}\rfloor$ in the exact expansion. Lemma 3.1 bounds the dominant excess by 95/147. For the opposite branch,

$$N^{(y+1)/2} \le (Q + t/16)^{(y+1)/4} < 2Q^{(y+1)/4},$$

because $(1 + t/(16Q))^{(y+1)/4} < (33/32)^2 < 2$. Hence

$$\left|\frac{M_t(s_-)}{M_t(s_+)}\right| \sum_{n=1}^{N} n^{(y-1)/2} W_n(s_-) < 4Q^{(1-y)/4} \le 4Q^{-1/2} < \frac{23}{500}. \tag{6.1}$$

The last inequality follows from $4\pi < 13$ and $16 \cdot 13/10^5 < (23/500)^2$. The normalized ordinary approximation errors are less than

$$\frac{242/147 + 23/500}{6000} < \frac{3}{10000}.$$

Lemma 4.1 applies because $T \geq 50000$. Its prefactor 2 and the carrier ratio bound both remainders by

$$4Q^{-(1+y)/4} \leq 4/Q < 52/x.$$

Thus

$$\begin{aligned} |H_t(x+iy)/M_t(s_+) - 1| &< \frac{95}{147} + \frac{23}{500} + \frac{3}{10000} + \frac{52}{100000} \\ &= \frac{5094127}{7350000} < \frac{347}{500} < \frac{7}{10}. \end{aligned}$$

For $x \geq 2 \cdot 10^6$, retain the sharper ordinary error $1/50000$ and the remainder prefactor 1 in (4.2). The opposite branch is then less than $11/1000$, since $16 \cdot 13/(2 \cdot 10^6) < (11/1000)^2$. The ordinary errors total less than $1/25000$, and the two remainders total at most $2/Q < 26/x$. Therefore

$$\begin{aligned} |H_t(x+iy)/M_t(s_+) - 1| &< \frac{95}{147} + \frac{11}{1000} + \frac{1}{25000} + \frac{26}{2 \cdot 10^6} \\ &= \frac{96624791}{147000000} < \frac{48}{73} < \frac{33}{50}. \end{aligned}$$

All estimates are uniform on their respective stated domains. □

7 From the comparison to a local count

Lemma 7.1 (A disk logarithmic-derivative estimate)**.** *Let F be holomorphic near a closed disk $|z-w| \leq r$ and satisfy $|F-1| \leq q$, where $r > 0$ and $0 < q < 1$. Then*

$$\left|\frac{F'(w)}{F(w)}\right| \leq \frac{2q}{r(1+\sqrt{1-q^2})}. \tag{7.1}$$

Proof. Set $f(\zeta) = (F(w+r\zeta)-1)/q$ and $a = f(0)$. Schwarz–Pick gives $|f'(0)| \leq 1-|a|^2$; for a constant map with $|a| = 1$, the same statement holds with derivative zero. With $s = |a|$,

$$\left|F'(w)/F(w)\right| \leq \frac{q}{r}\frac{1-s^2}{1-qs} \leq \frac{2q}{r(1+\sqrt{1-q^2})}.$$

The last maximum occurs at $s = q/(1+\sqrt{1-q^2})$, as follows by differentiation. All denominators are positive for $0 \leq s \leq 1$. □

Lemma 7.2 (Comparison-to-count extraction)**.** *Fix $0 < t \leq 1/2$ and $X \geq 200$, and put*

$$U = X + \tfrac{1}{2}, \qquad w = U + \tfrac{7}{2}i, \qquad F_t = H_t/B_t.$$

If $\sup_{|z-w|\leq 1/2} |F_t(z) - 1| \leq q$ for $0 < q < 1$, then

$$K_t(X) \leq \frac{2003}{2240}\log(X+2) + \frac{25}{7}\frac{4q}{1+\sqrt{1-q^2}}. \tag{7.2}$$

Proof. The disk has imaginary parts in $[3,4]$ and maps under $s = (1-iz)/2$ to real part at least 2. Thus B_t is nonzero and F_t is holomorphic near the closed disk. Lemma 7.1 applies with $r = 1/2$. At $s = (1-iw)/2 = 9/4 - iU/2$, let $L_U = \log(U/(4\pi))$. The exact formula for α gives

$$\begin{aligned} |\alpha(s)| &\leq L_U/2 + \pi/4 + 3/U + 81/(16U^2) < L_U/2 + 1 < \tfrac{1}{2}\log(X+2), \\ |\alpha'(s)| &\leq 1/U + 6/U^2 \leq 103/20000 < 3/500. \end{aligned}$$

Here $4\pi > e^2$ and $U < X + 2$. Holding time fixed,

$$\frac{B_t'}{B_t}(w) = -\frac{i}{2}\alpha(s)\left(1 + \frac{t}{2}\alpha'(s)\right), \qquad \left|B_t'/B_t\right|(w) \le \frac{2003}{8000}\log(X+2). \tag{7.3}$$

Evenness, order one and $H_t(0) > 0$ give the paired Hadamard identity [1, (105)]. Put $v = 7/2$. Its imaginary part is

$$D_t(U) := -\operatorname{Im}\frac{H_t'}{H_t}(w) = \sum_{\rho}\frac{v - \operatorname{Im}\rho}{(U - \operatorname{Re}\rho)^2 + (v - \operatorname{Im}\rho)^2}. \tag{7.4}$$

The scalar sum is positive and absolutely convergent by the strip bound and order-one growth, independently of the count being proved.

For a zero in the unit window put $h = U - \operatorname{Re}\rho$; then $|h| \le 1/2$. Set $p_h(a) = a/(a^2 + h^2)$. On $[5/2, 9/2]$,

$$p_h''(a) = \frac{2a(a^2 - 3h^2)}{(a^2 + h^2)^3} > 0.$$

The real source makes H_t real entire, so conjugate nonreal zeros have equal multiplicity and the same window membership. If their imaginary parts are $\pm b$, the strip bound $|b| \le 1$ and convexity give

$$\frac{p_h(v - b) + p_h(v + b)}{2} \ge p_h(v) \ge \frac{7/2}{(7/2)^2 + 1/4} = \frac{7}{25}.$$

Real zeros obey the same per-multiplicity inequality directly. Thus

$$K_t(X) \le \frac{25}{7}D_t(U). \tag{7.5}$$

Combining the disk estimate, (7.3) and (7.4) proves (7.2). Pairing preserves membership because the window restricts only real parts. Both walls satisfy the distance bound, so the specified half-open convention and multiple roots require no further exception. □

Proof of Theorem 1.1. For $X \ge 10^5$, the whole extraction disk lies in Theorem 1.3's domain. Use $q = 7/10$. The derivative allowance in (7.1) is less than $5/3$: for $0 \le s \le 1$, clearing its positive denominator reduces

$$\frac{7}{5}\frac{1 - s^2}{1 - (7/10)s} \le \frac{5}{3}$$

to $84s^2 - 70s + 16 = 84(s - 5/12)^2 + 17/12 > 0$. Lemma 7.2 consequently gives

$$K_t(X) \le \frac{2003}{2240}\log(X+2) + \frac{125}{21}.$$

Since $2003/2240 < 9/10$ and $125/21 < 6$, the rounded assertion follows. □

Proof of Corollary 1.2. For $X \ge 2 \cdot 10^6$, use $q_0 < 48/73$ from (1.8). The disk allowance is increasing in q, and at $q = 48/73$ its square root is $55/73$ and

$$\frac{4q}{1 + \sqrt{1 - q^2}} = \frac{3}{2}.$$

The additive count allowance is therefore $75/14 < 11/2$. □

8 Scope and limitations

This is an explicit upper count on a restricted high-height domain. An upper count alone supplies neither a lower population, an exact interior integer count, nor a lower bound on a derivative at a zero. The theorem is compatible with multiple zeros and collisions. The paired kernel argument applies to any real entire function meeting its Hadamard and strip hypotheses; the complete theta expansion is the additional input used to obtain the particular numerical comparison. Positivity, evenness or reflection of a modified source does not supply that expansion.

In particular, the Euler-factor, weighted-square, reflected and positive-primitive control families recorded in the research repository do not challenge a multiplicity-count upper bound merely by colliding. No untested control is claimed to fail (1.7) on a particular box. No noncollision conclusion is drawn.

The lower-height gap is stated rather than filled by compactness. The local argument does not provide an evaluated cumulative estimate for $N_t(X)$. If a later application uses Polymath's cumulative comparison function, the normalization to retain is

$$g(X,t) = \frac{X}{4\pi}\log\frac{X}{4\pi} - \frac{X}{4\pi} + \frac{11}{8} + \frac{t}{16}\log\frac{X}{4\pi}.$$

The heat drift and the constant $11/8$ should not be replaced by a time-zero or averaged convention. A conductor-dependent version for Dirichlet families also requires additional estimates.

The remaining height gap is $200 \le X < 10^5$. The bounded continuation stops at its present result, pending cross-model review of the changed cutoff and extraction. A separate application must specify its kernel, domain, baseline and numerical tolerance before further optimization or a gain assessment. In particular, a principal-value velocity tail needs symmetric cancellation and two-sided cumulative information; an upper local count alone does not provide these. The draft makes no assertion that the bound improves an existing certificate.

A Source normalization and reproducibility

The source statements actually used are identified in Section 2. In the proof following (99) of [1], the displayed logarithmic contribution to $\operatorname{Re} s_*$ uses $t/2$, while its definitions (9), (17) and its explicit estimate (21) give $t/4$. We avoid that display and use the exact identity (3.4). This observation does not dispute the source's time-uniform asymptotic theorem.

Immediately before (60) of [1], the printed inequality is

$$\frac{1}{6(T'-.66)} \le \frac{1/3}{4(T-3)}.$$

It fails by a factor tending to two. Replacing the numerator $1/3$ by $2/3$ gives the valid bound used in (4.3). Our envelope $(u^2+1)/D$ is conservative. Also, the displayed tail for (63) can omit the remainder when the height index wins: its upper index range may be empty although RS_m is still present. The separate E_a in (4.6) retains it. These small source slips do not defeat Polymath's conclusions.

The standard-library arithmetic programs in `../numerics/` are:

- `check_local_count_reduction.py`: the original kernel, carrier chain rule and extraction constants.
- `check_off_axis_comparison.py`: the original high-height comparison budget and Gaussian/remainder allowances.

- `round-2/check_lower_cutoff_and_paired_extraction.py`: the reduced cutoff, retained full remainder denominator, Schwarz–Pick constants and conjugate-pair kernel.

Their small JSON records identify the code and note hashes. The round-two record is `../numerics/round-2/LOWER_CUTOFF_AND_PAIRED_EXTRACTION_CHECKS_20261017.json`. The older remainder checker now labels its endpoint equality as an equality; analytic strictness is supplied upstream.

Each normal run exits zero. The original failure injections remain `--kernel-floor 9/40` and `--comparison-ceiling 13/20`. The round-two injections are `--comparison-ceiling 69/100`, `--kernel-floor 3/10`, and `--derivative-allowance 3/2`; each exits one. The paired kernel floor is attained by a real zero at horizontal distance $1/2$, including a double root counted twice. For the disk estimate, with $r = 1/2$,

$$F(w + r\zeta) = 1 + \frac{7}{10}\frac{\zeta - 2/5}{1 - (2/5)\zeta} \quad \Longrightarrow \quad |F'(w)/F(w)| = 49/30 > 3/2.$$

This is an exact disk-automorphism test, not a sample of H_t. The comparison-ceiling injection fails a sufficient budget, not the actual heat comparison. The separate class extremizers are not claimed to be realized simultaneously by theta.

Finite arithmetic and class tests do not formally verify the classical imports, contour shifts or infinite-domain analytic arguments. Their purpose is authoring support. The current build record also distinguishes Note 01's pre-existing single-link repair from its historical reviewed hash; reversing that repair recovers the reviewed bytes. The notes and reviews retain their source identities, and the earlier manuscript is recoverable from commit `146b3f9`, without a new snapshot folder.

Acknowledgment of LLM assistance

This draft was prepared for Edward Baker with substantial assistance from OpenAI's `gpt-6.1-sol` model in Codex, using reasoning effort `ultra`, on 17 October 2026. The assistance included derivation, source comparison, same-model development work, numerical bookkeeping, manuscript drafting and compilation checks. The research notes record these contributions and their limitations. The baseline received Claude cross-model LLM review; cross-model review of the round-two extension and specialist human review remain pending. Responsibility for checking, revising and endorsing any eventual submission remains with its human author.

References

[1] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Research in the Mathematical Sciences **6**, article 31 (2019). doi:10.1007/s40687-019-0193-1. The source numbering used here follows arXiv:1904.12438v2, 4 August 2019.

[2] C. Bellotti and P.-J. Wong, *Improved estimates for the argument and zero-counting function of the Riemann zeta-function*, arXiv:2412.15470v2, 7 July 2025, Corollary 1.6. Used here only for a time-zero benchmark.